Elasticity and Its Application

Reference: Microeconomics by Mankiw (8e)

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Elasticity of Demand

- **Price elasticity of demand.** How quantity demanded responds to change in price.
- Inelastic: Only slight change in change in demand for a change in price.
- Even without going into details, it is evident that this is quite important. Why?

Some Reasons

- Availability of Close Substitutes.
- Necessities versus Luxuries
- Definition of the Market (This is a little bit of a digression).
- Time Horizon. Again, we are not studying dynamic models so this is a little different, but it is still worth thinking about.

Formal Definition

$$\text{price elasticity of demand} = \frac{\text{percentage change in quantity demanded}}{\text{percentage change in price}}$$

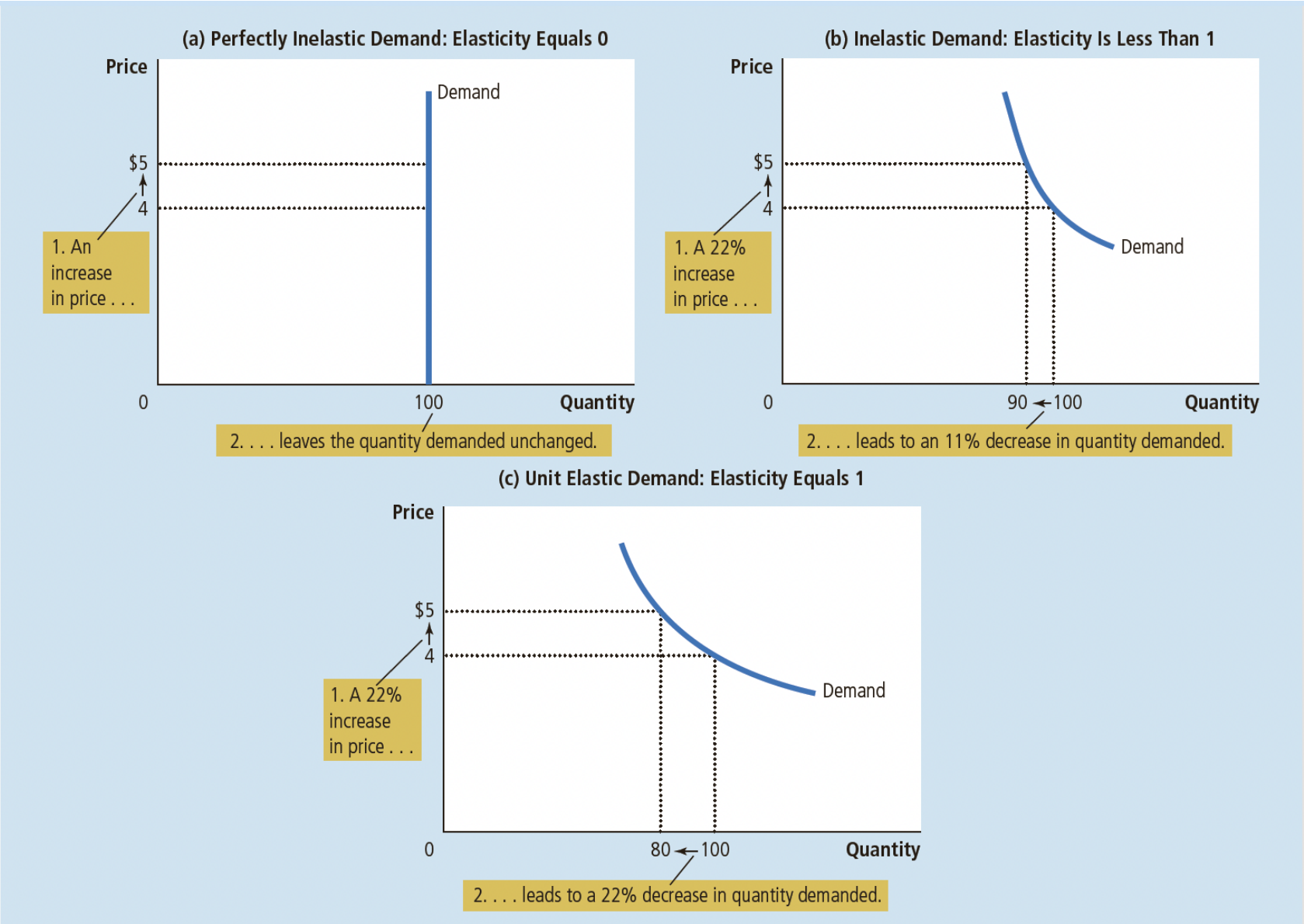
- Some issues: You get a different elasticity when going from A to B vs B to A.
- Price goes from \$6 ($q = 120$) to \$4 ($q = 80$) vs vice versa. We get different values for elasticity.

Mid-Point Method

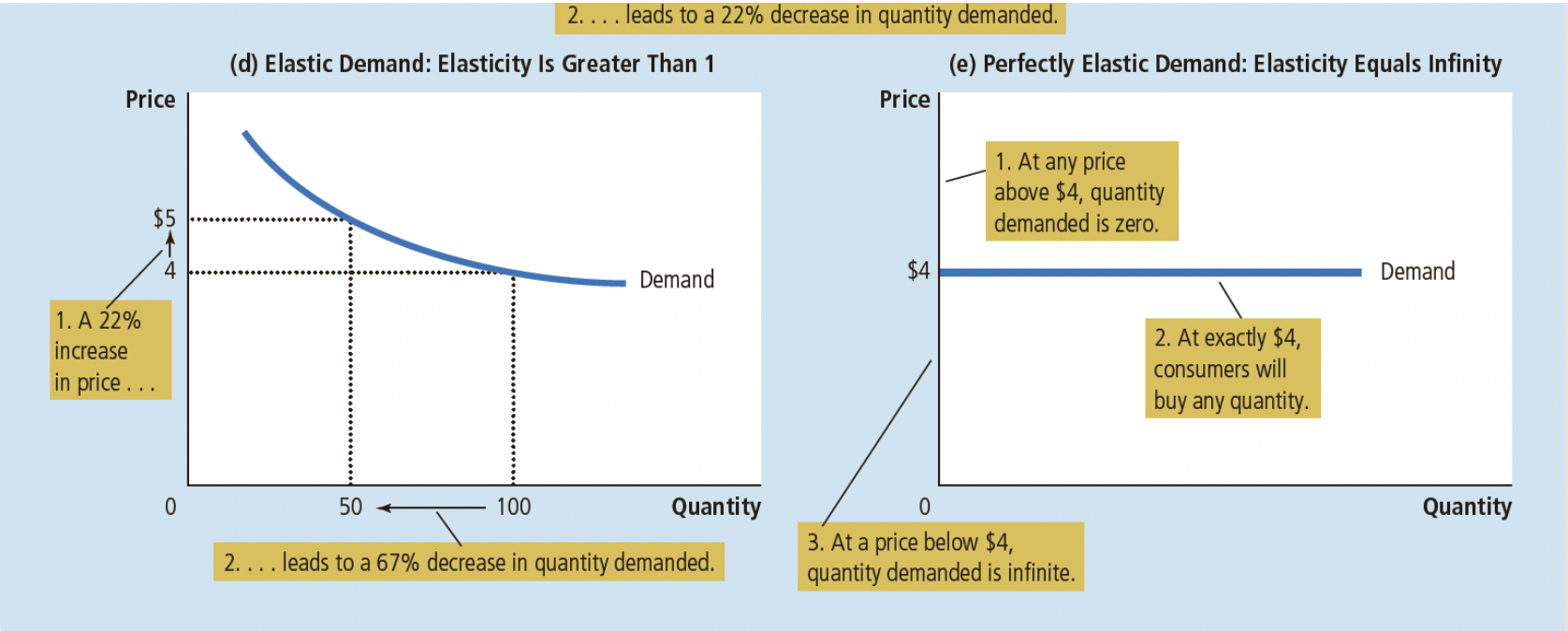
$$\text{price elasticity of demand} = \frac{\frac{Q_1 - Q_2}{\left(\frac{Q_1 + Q_2}{2}\right)}}{\frac{P_1 - P_2}{\left(\frac{P_1 + P_2}{2}\right)}}$$

- This book doesn't often engage in such calculations. But we will when needed.

Graphical Representation



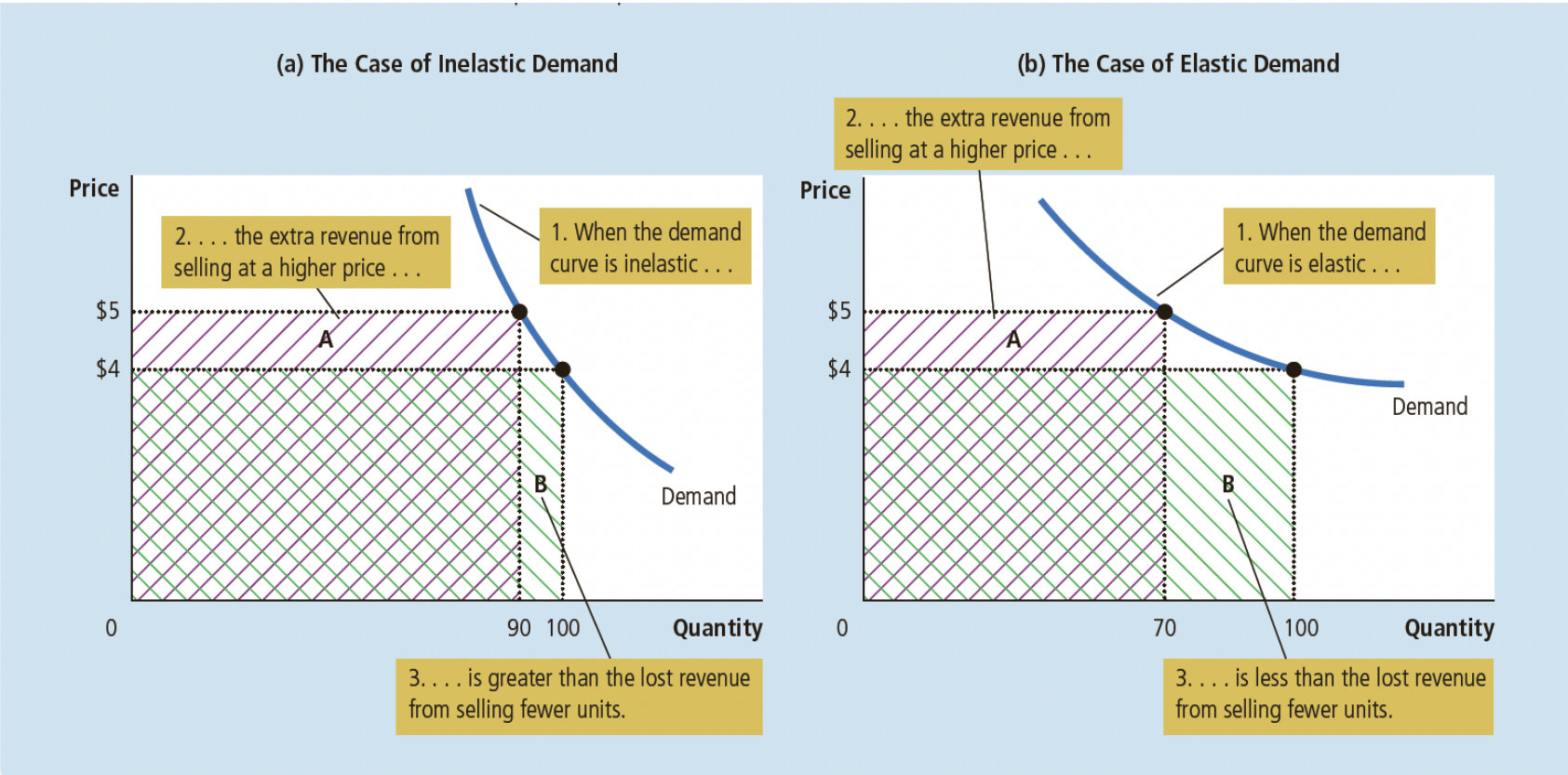
Graphical Representation



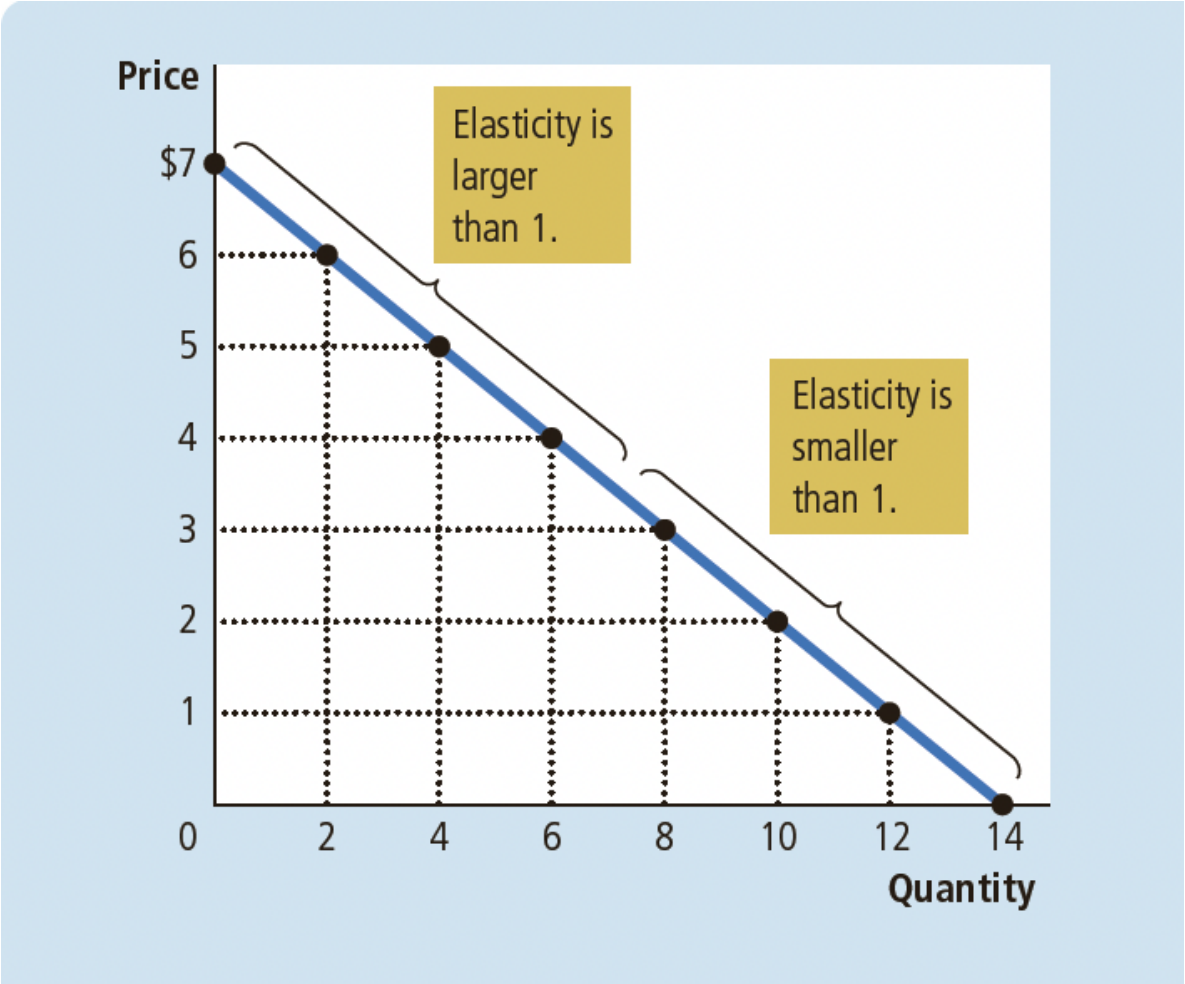
Revenue

- How to think about revenue in this context?
- It is clear that the elasticity of demand is what drives how changes in price affects revenue as well.
- Inelasticity means people find it hard to stop buying a good even for higher prices.

Graphical Representation



Counter Intuitive Result



Other Demand Elasticities

$$\text{income elasticity of demand} = \frac{\text{percentage change in quantity demanded}}{\text{percentage change in income}}$$

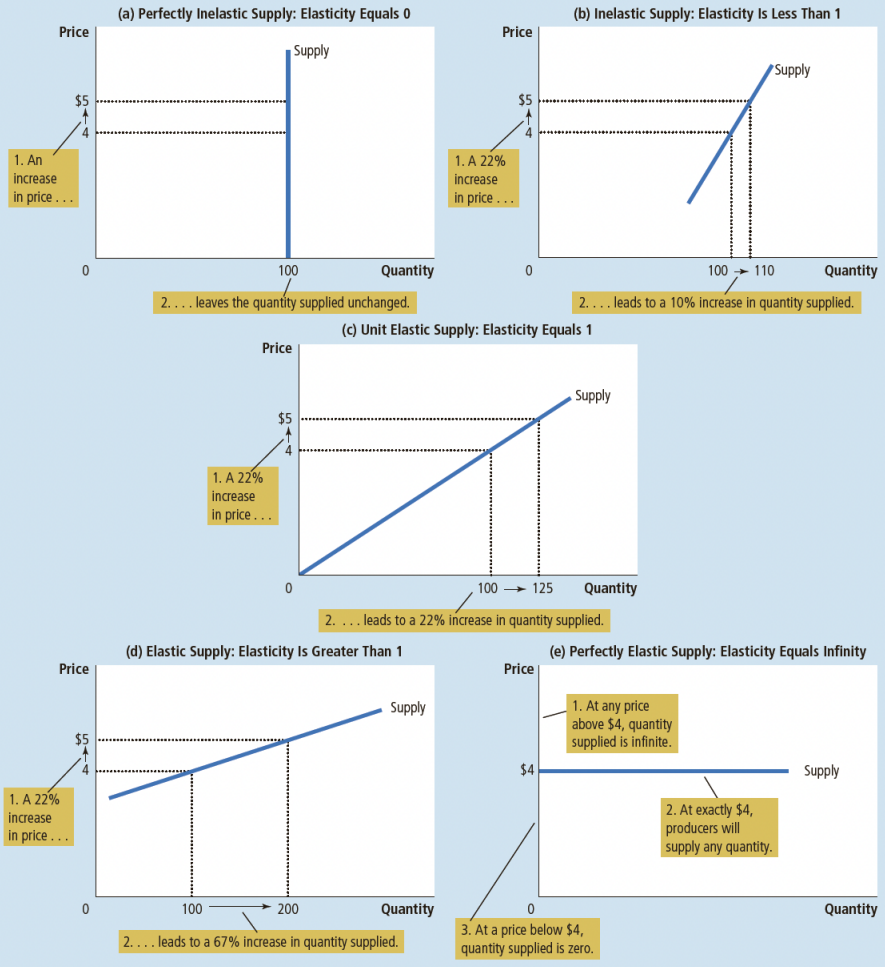
$$\text{cross-price elasticity of demand} = \frac{\text{percentage change in quantity demanded of good 1}}{\text{percentage change in the price of good 2}}$$

Price Elasticity of Supply

$$\text{price elasticity of supply} = \frac{\text{percentage change in quantity supplied}}{\text{percentage change in price}}$$

Graphical Representation

The price elasticity of supply determines whether the supply curve is steep or flat. Note that all percentage changes are calculated using the midpoint method.

FIGURE 5

The Price Elasticity of Supply